Shaking Faster Does Not Make The Wave Go Faster

SYNOPSIS

In Note 1 you shook a slinky and watched a disturbance walk away from your hand while the ribbon stayed home. What you could not do was *measure* it. A stretched slinky will not hold still long enough to be measured — the pulse runs to your partner, bounces, comes back, and the whole thing is over in a second or two. So Note 1 stopped at "there is a repeat" and never put a number on it.

This note supplies the numbers, and it turns out there are only three of them. A moving wave repeats in **both space and time**. Freeze the motion — take a snapshot — and the repeat is laid out along the string as a *distance*: that is the **wavelength**, λ. Or stand still at one spot and let the wave come at you, and the repeat arrives over and over as a *timing*: that is the **period**, *T*, and its partner the **frequency**, *f*. Add how far the string swings from its rest position — the **amplitude** — and the wave is described completely. Everything else follows from those three, including the one thing most people guess wrong: how fast it travels.

ABOUT YOUR SLINKY

You are about to get those numbers out of a simulation instead of a slinky, and it is worth being honest about why. In Note 1 *both ends of your slinky were held* — you held one, your partner held the other. That is why your pulse ran to the far end, bounced, and came back at you, and why everything was over in a second. That is a **practical limitation, and an unavoidable one** — not physics. What we actually want is an **infinite slinky**: one that runs on forever, so a wave leaves your hand and never returns to muddle what you are watching. Nobody can buy one. The simulation can build one: set the far end to **No End** and the string runs off the screen forever, and set **Damping** to **None** so nothing drains the wave's energy. So do not think of the simulation as a tidier slinky. Think of it as *the slinky you could not build* — the ideal your real one was trying to be.

Performance Expectations

Describe a wave with three numbers — amplitude, wavelength and frequency — and generate waves with different values of each using the simulation's controls. Measure a wavelength and a period from the simulation and combine them into the wave's speed. Predict what changing the frequency, the amplitude and the tension does to that speed, and check every prediction against the simulation.

Preamble — set the simulation up

Get the simulation running and matched to the settings below before you answer anything. Every question after this one assumes your screen looks like Fig. 1.

You will need: A computer or tablet with a web browser, this handout and a pencil. No slinky today — the simulation is the slinky you could not build. The simulation is free, needs no account and installs nothing.

Run the simulation: `phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_en.html`
If that ever fails, search for *"wave on a string phet"* — PhET call it *Wave* on a String, singular.

1. Open the simulation in its own browser tab so you can go back and forth between it and this handout.
2. Choose **Oscillate** on the left, **No End** on the right, and **Normal** for the playback speed.

3. Slide **Damping** to **None** and **Tension** to the middle (**Medium**).
4. Check the boxes for **Rulers**, **Timer** and **Reference Line**.
5. Set **Amplitude** to **0.50 cm** and **Frequency** to **1.00 Hz**.
6. Check your screen against Fig. 1. You can drag the rulers wherever you want them.

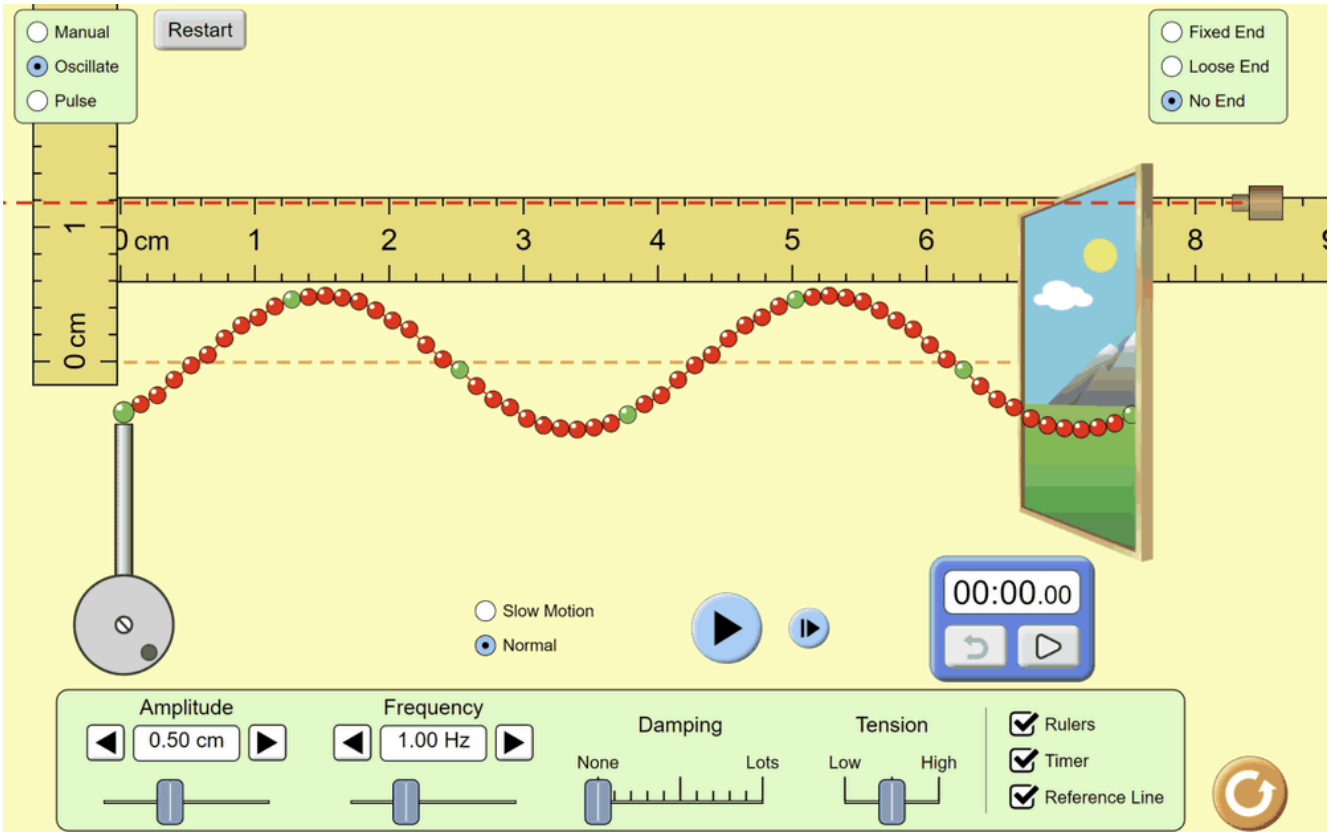


Fig 1. The simulation with the settings above, paused at `00:00.00`. The wrench at the far right is the **No End** setting — the string carries on past the edge of the picture and nothing comes back. The dashed line across the middle is the **Reference Line**. Notice the **green beads** spaced along the string: those are the simulation's version of the twist-tie you put on your slinky in Note 1, and Question 2 asks you to watch one.

Instructions

Work through these with the simulation open in front of you. Where a question asks you to predict something, write the prediction down *before* you move the slider — being wrong on paper is how this note teaches.

1. Press **Play** to start the wave. *How is the moving wave generated?* Look at the left-hand end of the string and describe what the simulation is doing there. Then say in one sentence how it compares with what your own hand did in Note 1.

2 Now watch *one bead*. Pick a single green bead, keep your eye on that bead alone, and follow it for several seconds while the wave marches steadily to the right. **Which way does your bead move?** How far does it ever get from where it started? And what did the twist-tie on your slinky do in Note 1 — the same thing, or something different?

ONE THING THIS SIMULATION CANNOT SHOW YOU

Your bead goes up and down; the wave goes sideways. Those are at right angles, so every wave in this simulation is a **transverse** wave — Fig. 2 of Note 1, running on a screen. The simulation has no way of making the *other* kind. Sound, the wave this whole course is about, is **longitudinal**, and you will not find it here. That is a limitation of the simulation, not of the physics: amplitude, wavelength and frequency describe a longitudinal wave exactly as well, and everything you measure in this note carries straight over to sound.

3 What does the **Reference Line** (the dashed line) indicate?

4 Slide the **amplitude** back and forth, then set it back to 0.50 cm. What feature of the wave does it change? Measured from where, to where — what is the amplitude measured *in reference to*?

5 **Freeze the wave in time** by pressing pause. Frozen, the wave gives up its *spatial* features to the ruler. A wave's **wavelength** is the peak-to-peak distance, the trough-to-trough distance, or the **length of one full repeat** of the wave. Measure it.

$\lambda =$ _____ cm

6 Press **Play** to let the wave move again. Which way is the wave moving?

7 Imagine yourself standing at the **3 cm** mark on the horizontal ruler, facing the waves coming at you. Start the **Timer** as a peak passes you, and stop it when you have counted the *next ten* peaks. How long did the ten peaks take? Therefore, how long did it take to go from one peak to the next?

8 Repeat Question 7 with the frequency set to 1.5 Hz, 2 Hz, 2.5 Hz and 3 Hz. How long did it take to count 10 peaks after starting the Timer? Use the **Slow Motion** option if the waves are moving too fast to count.

	Time for 10 peaks to pass	Time for one peak to pass
1.5 Hz		
2 Hz		
2.5 Hz		
3 Hz		

9 Look down your two columns. Determine the relationship between the **frequency** (*f*, lowercase) and the time taken for one peak to pass. Hint: the unit Hz is equal to 1/s.

10 The time it takes to go from one peak to the next is the wave's **period, *T*** (capital T). Note that the period is **NOT** the same thing as the wavelength — the period is a *time*, the wavelength is a *distance*. The period is the time for one full repeat of the wave. Write the relationship between *f* and *T* both ways round. **Don't forget units.**

f = _____

T = _____

11 The frequency *f* is the number of peaks you count in one second — the **number of repeats (cycles) in one second**. If you count 10 peaks in 5 seconds, how many repeats is that in 1 second? What is the frequency in Hz? Does it agree with your table above?

12 If your wave has a frequency of **15 Hz**, how many full repeats (cycles) are there in one second? How long does one full repeat take? What is the period of the wave? **Don't forget units.**

f = _____

T = _____

13 Moving waves clearly have speeds. The **speed** of a wave is simply how fast one full cycle runs past you if you stand at a fixed position and watch the wave go by: one wavelength, in one period. So the speed v (in m/s) is $v = \lambda / T$ — the wavelength (m) divided by the period (s). **Show that this is the same as saying the speed is frequency times wavelength.**

14 Do you think that changing the frequency changes the *speed* of the wave? **Write down your prediction before you touch anything.**

Now check it: set each frequency, freeze the wave, measure the wavelength off the ruler, and work out the speed.

Frequency (Hz)	Wavelength (cm)	Speed (cm/s)
1		
2		
3		

Speed = frequency × wavelength

Conclusion:

15 Should changing the **amplitude** change the speed? Hint: does the equation for the speed contain the amplitude? Confirm your answer with the simulation.

16 Does the **Tension** affect the speed? If so, how? Change it and watch.

17 Slide the **Damping** up from None. What does damping do? Which feature of the wave does it affect? And — thinking back to Note 1 — which real-world thing does turning damping up put *back into* the simulation?

18 Choose the **Pulse** option and press the green button to send a single “pulse” along the string. **Is it a wave?** If so, how? If not, why not?

Check yourself

These last eight are quick ones — no simulation needed. Underline your answer for each situation. Figures 2 and 3 show a rope with a knot tied at the point marked **A**; in both, the hand is at the left and the wave is travelling to the right.

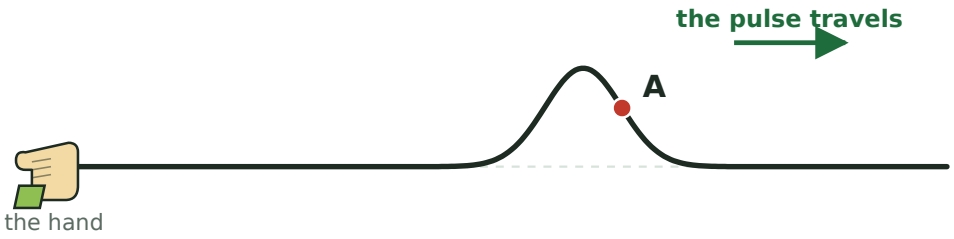


Fig 2. A single pulse, generated by one flick of the hand at the left, part way along a long rope. The knot at **A** sits on the leading side of the pulse. Questions 1–4 refer to this picture.

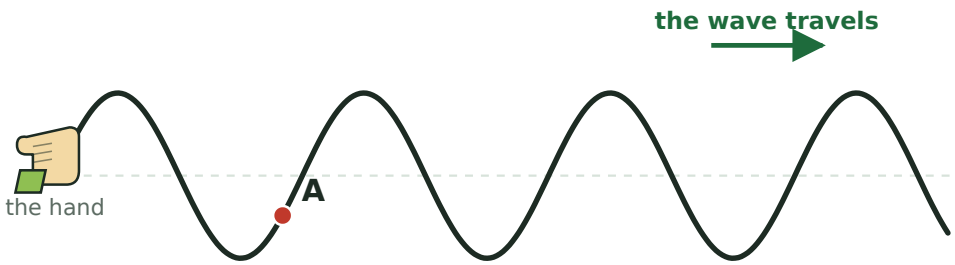


Fig 3. The same rope, with the hand now shaking up and down over and over so a continuous wave runs to the right. The knot at **A** is again marked on the rope. Questions 5–7 refer to this picture.

1. If you advance the movie one frame, the knot at point A would be
   a) in the same place **b)** higher **c)** lower **d)** to the right **e)** to the left
2. If the person generates a new pulse like the first but more quickly, the pulse would be
   a) same size **b)** wider **c)** narrower
3. If the person generates another pulse like the first but he moves his hand further, the pulse would be
   a) same size **b)** taller **c)** shorter
4. If the person generates another pulse like the first but the rope is tightened, the pulse will move
   a) at the same rate **b)** faster **c)** slower
5. If you advance the movie one frame, the knot at point A would be
   a) in the same place **b)** higher **c)** lower **d)** to the right **e)** to the left
6. If you advance the movie one frame, the pattern of the waves will be _____ relative to the hand.
   a) in the same place **b)** shifted right **c)** shifted left **d)** shifted up **e)** shifted down
7. If the person starts over and moves his hand more quickly, the peaks of the waves will be
   a) the same distance apart **b)** further apart **c)** closer together
8. If you lower the frequency of a wave on a string you will
   a) lower its speed **b)** increase its wavelength **c)** lower its amplitude **d)** shorten its period

WHERE THIS IS GOING

Question 14 is the one worth carrying out of this note. Shaking faster did not make the wave go faster — it packed the same speed into shorter wavelengths. Tightening the string *did*. The speed of a wave is set by **the medium it travels in**, not by whoever is shaking the end. Now think about what that means for a room full of sound: high notes and low notes are shaken at wildly different frequencies, but they all travel through the same air, so they all arrive at the same speed. Strike a chord and it reaches you as a chord — not smeared out with the high notes first. That is the medium doing its job, and it is where we go next.

Copyright © 2026 by Dr. Byungmoon Cho · FLCC Physics · PHY 105